

MALAV GAJERA

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Education

Northeastern University

Boston, MA

Master of Science in Computer Software Engineering, GPA: 3.9

Sept 2024 – Dec 2026 (Expected)

- **Relevant Coursework:** Object-Oriented Design, Data Structures and Algorithms, Cloud Computing, High-Performance Parallel ML & AI, Generative AI

Nirma University

Ahmedabad, Gujarat

Bachelors of Technology in Computer Science and Engineering

Oct 2020 – May 2024

Technical Skills

Languages: Python, TypeScript, Java, JavaScript (ES6+), C#, SQL, C/C++, Go, Bash

Backend & Frameworks: Django, FastAPI, Spring Boot, ASP.NET Core, Entity Framework, REST APIs, gRPC, GraphQL, WebSockets, Microservices, JWT/OAuth2, Celery, Hibernate, RabbitMQ, Kafka, Message Queues

Frontend: React, Next.js, Angular, HTML5, CSS3, Tailwind CSS, Redux, Responsive Design

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch, Query Optimization, Indexing Strategies, Schema Design

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch, IAM, VPC, ELB, ECS Fargate, ElastiCache, SQS), GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Azure DevOps

Monitoring & Observability: Prometheus, Grafana, Datadog, Structured Logging, Distributed Tracing, ELK Stack

Testing & Tools: Pytest, Jest, xUnit, Mockito, Playwright, Postman, Swagger/OpenAPI, Git, JIRA, Agile/Scrum

Experience

Crewasis

Jan 2026 – May 2026

Software Engineer Co-op

New York, USA

- Accelerated API response times by **35%** and cut database load by **30%** through **PostgreSQL** query profiling (index tuning, ORM batching) and **Redis** caching layer, sustaining sub-200ms p99 responses under peak traffic for **5K+ monthly active users**
- Reduced release cycle duration by **40%** by containerizing services with **Docker** and orchestrating end-to-end **CI/CD** pipelines in **GitHub Actions**; provisioned **AWS** infrastructure (EC2, RDS, S3, Lambda) with **Terraform** IaC enabling zero-downtime blue-green rollouts across staging and production environments with instant rollback and canary monitoring
- Delivered production backend services in **Python (Django)** and **React**, designing JWT-authenticated **REST APIs** with input validation and structured error handling; maintained **90%+ test coverage** (Pytest/Jest) across weekly shipments

Tatvasoft

Jan – May 2024

Software Developer Intern

Ahmedabad, Gujarat

- Developed **110+ RESTful API** endpoints across 8+ domain modules using **C#/ASP.NET Core** and **Entity Framework**, supporting 100+ concurrent users with sub-500ms latency through async controller patterns and service-layer architecture
- Boosted database throughput by **30%** via **PostgreSQL** query tuning, composite indexing strategies, and Repository Pattern refactoring of ORM interactions across 15+ high-traffic tables, cutting average query execution time by **45%**
- Automated **CI/CD** workflows in **Azure DevOps** and strengthened test reliability with **xUnit**-based unit and integration suites, cutting production defects by **20%** and bolstering release confidence across sprint cycles

Projects

DISTRIBUTED TASK QUEUE | FastAPI, Celery, RabbitMQ, Redis, Prometheus, AWS

- Engineered production-grade distributed task queue processing asynchronous workloads (image transformation, PDF rendering, webhook dispatch) through **FastAPI**, **Celery**, and **RabbitMQ**, implementing exponential-backoff retry logic with dead-letter routing and configurable priority scheduling that achieved zero message loss across **10K+** queued jobs
- Designed resilience layer with sliding-window rate limiting, **Redis**-backed circuit breakers, and idempotency-key deduplication — eliminated **100%** duplicate executions while sustaining sub-50ms API latency under concurrent load
- Deployed **Prometheus/Grafana** observability stack with correlation-ID tracing and latency dashboards; containerized 8 microservices via **Docker** Compose and codified **AWS** infrastructure (ECS Fargate, RDS, ElastiCache) using **Terraform**

RESILIENT CLOUD DEPLOYMENT PLATFORM | AWS, Terraform, FastAPI, Docker, GitHub Actions

- Architected fault-tolerant **AWS** infrastructure handling **10K+ daily requests** using EC2, RDS, S3, Lambda, and SNS/SQS with **Terraform** IaC; enforced multi-AZ distribution, VPC segmentation, and NAT gateway routing for high availability
- Streamlined **FastAPI** backend delivery with **Docker** containerization and **GitHub Actions CI/CD**, enabling seamless blue-green rollouts with continuous health checks, rollback triggers, and real-time **CloudWatch** observability
- Configured auto-scaling architecture via **Elastic Load Balancer** with **IAM**-based access controls, security groups, and SSL termination, ensuring secure service-to-service communication and **99.9%** uptime under variable traffic loads

FULL-STACK TRANSPORTATION PLATFORM | Java, Spring Boot, PostgreSQL, React, Python

- Built microservices backend using **Java Spring Boot** with **30+ REST APIs** secured via **JWT** and **RBAC**, achieving sub-200ms response times through connection pooling, query tuning, lazy-loaded entity graphs, and paginated result sets
- Modeled **PostgreSQL** schema with caching strategies and optimistic locking to handle concurrent ride-booking workflows, strengthening transactional consistency and reducing request overhead by **25%** under concurrent booking loads
- Integrated NLP-driven support triaging module in **Python**, reducing manual ticket volume by **40%** and accelerating customer resolution through sentiment analysis, intelligent priority routing, escalation workflows, and SLA tracking